

Jay Nasriwala

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EDUCATION

- **Sarvajanik College of Engineering and Technology** Surat, India
Bachelor of Technology - Computer Engineering; CGPA: 9.38 2022 - 2026

WORK EXPERIENCE

- **Trainee Software Developer:** STPL Global Surat, India
Research & Development Department Jan 2026 – Present
 - Engineered a **photogrammetry-based 3D reconstruction pipeline**, integrating **U2-Net segmentation** to achieve **~95% masking accuracy** for automated extraction from **100+ multi-view images** per scan.
 - Created a **Structure-from-Motion (SfM)** pipeline with **OpenCV** for feature extraction, matching and sparse reconstruction generating **10,000+ points**, scaling to **1,00,000+** via dense reconstruction with **OpenMVG**.
 - Optimized **initial pose estimation** by enhancing **RANSAC based image pair selection** using **SQLite driven querying** to identify **top-10 image pairs with maximum inliers**, improving reconstruction stability and achieving **~96% higher accuracy** in model initialization.
 - Calibrated **IDS industrial camera** by tuning imaging parameters: **gains, exposure, and aperture**, and performed **geometric calibration** using a checkerboard pattern to compute the **camera intrinsic matrix**.
- **Software Developer Intern:** STPL Global Surat, India
Research & Development Department Jan 2026 - May 2026
 - Contributed to the **SlysX** project, developing a high-performance **3D geometry analysis and slicer engine** using **Python, Delphi, and DirectX 9**, with support generation, thickness and overhang analysis, auto-orientation, peel-force estimation, **Octree-based spatial partitioning**, ray casting, and parallel processing for meshes up to **1M triangles**.

PROJECTS

- **InsightAI - AI-Powered Research Assistant**
 - Architected a **Streamlit app** leveraging **Multi-Agent AI** using **CrewAI** and **Model Context Protocol (MCP)**, automating end-to-end research workflows using **2 specialized agents**.
 - Designed **2 custom MCP servers** exposing **3 tools** (web search, news search, image generation) by integrating **DuckDuckGo** and **Hugging Face FLUX.1** on a fully **free open-source stack**.
 - Configured **LLM (Llama 3.3 70B)** with **LiteLLM**, established **3-attempt retry logic**, subprocess agent orchestration, and **Pydantic** output parsing. [\(Source Code\)](#)
- **ResuMateAI - AI-Powered ATS Resume Optimizer**
 - Built a scalable **Streamlit-based AI system** leveraging **LLMs (Google Gemini)** to analyze job descriptions and resumes, delivering **ATS match scoring** and **actionable optimization insights** in real time.
 - Designed end-to-end **resume pipeline** including **skills gap analysis**, **ATS compliance checks**, and **cover letter generation**, improving alignment with job requirements.
 - Engineered robust **document processing workflows** using **PyPDF2**, enabling accurate text extraction and dynamic PDF generation for production-ready outputs. [\(Source Code\)](#)

TECHNICAL SKILLS

- **Languages:** Python, C, Java, SQL, PL/SQL
- **Databases:** MySQL
- **Web Technologies:** HTML & CSS, React, Flask, FastAPI, REST API
- **Data Analytics & ML:** Machine Learning, Deep Learning, NLP, Statistics, Power BI, Tableau, MS-Excel, Advanced Excel
- **AI & Agentic Systems:** MCP (Model Context Protocol), Agentic AI, RAG Pipelines, CrewAI, LangChain, LangGraph

RESEARCH PAPERS & CERTIFICATIONS

- Malaria Parasite Detection in Thin Blood Smear Images Using ConvNeXt with Explanations [\(View Paper\)](#)
- Segmenting Brain Tumor from MRI using an Efficient Encoder-Decoder-based DeepLabv3+ Architecture [\(View Paper\)](#)
- Fundamentals of Deep Learning By NVIDIA [\(View Certificate\)](#)
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional [\(View Badge\)](#)